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BRAC UNIVERSITY
Department of Computer Science Engineering
Mid-Term Examination, Summer 2024
CSE 350: Digital Electronics and Pulse Techniques

Total Marks: **60 (20 x 3)**
Time: **90 Minutes**

- Total no. of questions are **four (4)**. Answer any **three (3)**.
- Figure in bracket [ ] next to each question indicates marks for that question.
- Use $V_\gamma(diode) = 0.6\ V$, $V_\gamma(transistor) = 0.5\ V$, $V_{BE}(forward\ active) = 0.7\ V$ $V_D(conducting\ voltage\ of\ Diode) = 0.7\ V$, for all the questions unless otherwise stated.

Name:	ID:	Section:

Question 1 [20]

For the given TTL circuit in Figure 1, $\beta_F = 30$, $V_{BE(sat)} = 0.8V$, $V_{CE(sat)} = 0.2V$, and $\beta_R = 0.1$. Assume **logical High** is $\mathbf{10V}$ and **logical Low** is $\mathbf{0.2V}$.

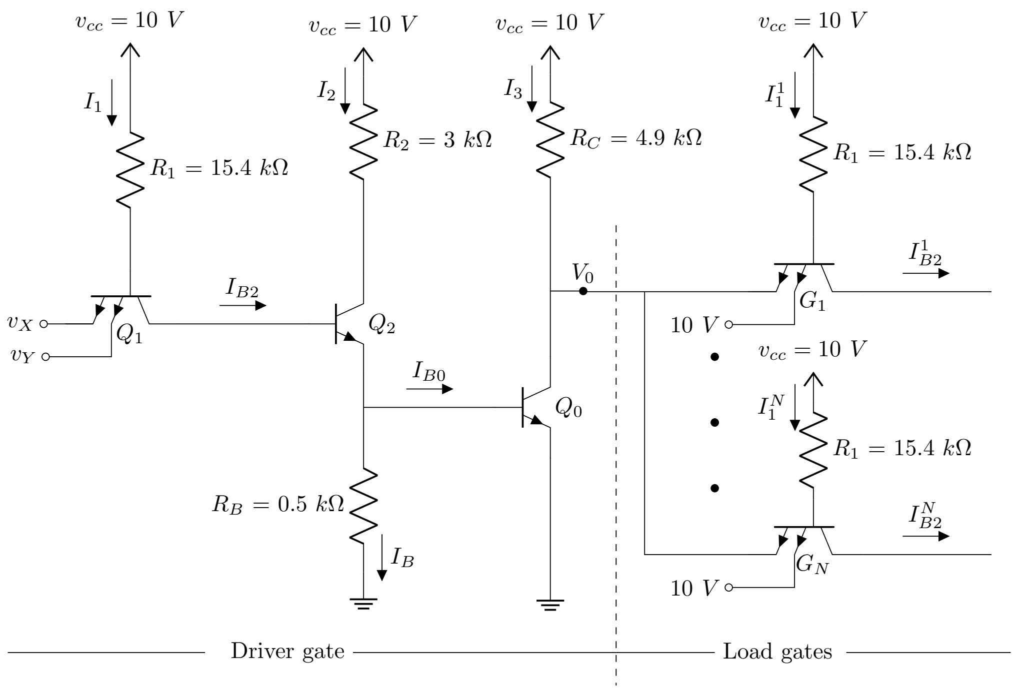

Figure 1

CO1	(a)	**Determine** the minimum amount of voltage required at the collector of Q_1 to turn ON both transistors Q_2 and Q_0 simultaneously.	[2]
CO1	(b)	**Calculate** the minimum value of β_{min} for transistors Q_2 and Q_0 such that this TTL logic gate can still function normally.	[10]
CO1	(c)	If 20 identical load gates similar to the given TTL circuit are connected to the output, **calculate** the output voltage when both inputs of the driver circuit are **Low**.	[8]

Question 2 **[20]**

Consider the diode-logic circuit in figure 2. Assume V(Logic 1)=5V and V(Logic 0)=0.2V.

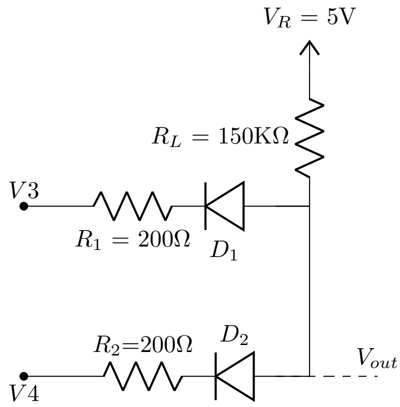

Figure 2: DL AND gate

$V_A\ (V)$	$V_B\ (V)$	$D1\ [on/off]$	$D2\ [on/off]$	$V_0\ (V)$

Table 1: Table for question 3

CO1	(a)	**Find** the output voltage and V_0 for all the input combinations and fill out the table above. Hence, **identify** the gate from the table.	[7]

Suppose we replace the resistors in the above circuit. Assume the output voltage V_0 from the table is fixed for $V_A = V_B = 0.2V$.

CO1	(b)	If the value of the maximum current is **doubled**, **find** the new resistor values.	[8]
CO1	(c)	**Calculate** the minimum value of R_L in kΩ if we want to keep the maximum power dissipation under 0.5 mW.	[5]

Question 3 [20]

For the transistors in the DTL & RTL circuit in figure 3 and 4 assume $\beta_F = 65$, $V_{BE(saturation)} = 0.8V$, and $V_{CE(saturation)} = 0.2V$. The input voltages A and B can be $0V$ or $5V$.

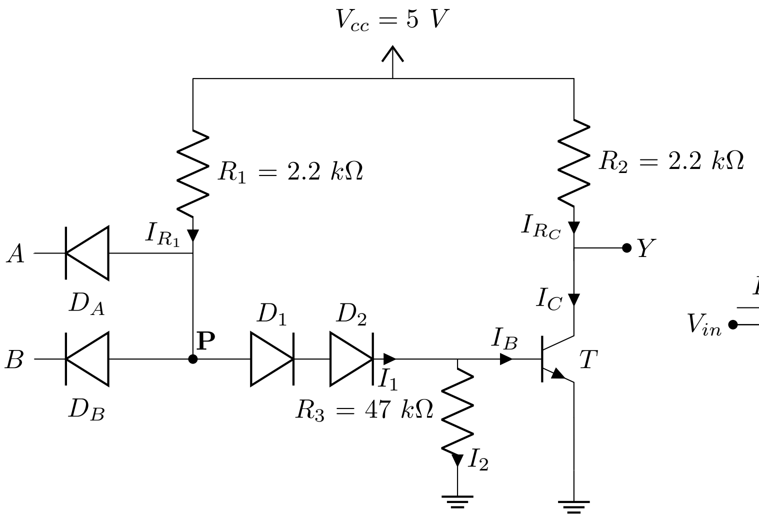

Figure 3: DTL NAND gate

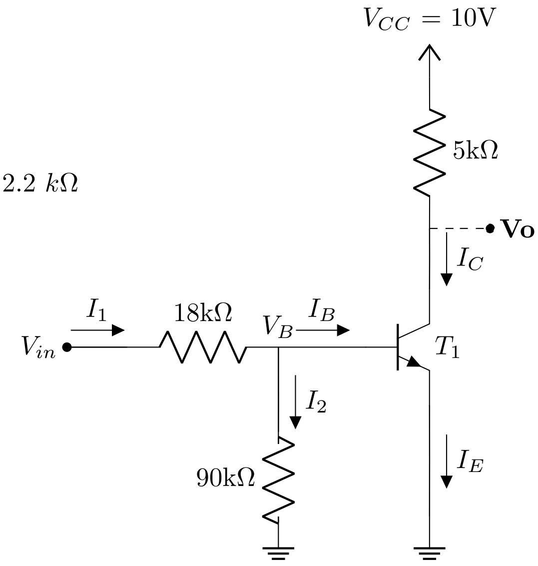

Figure 4: RTL for part (c) only

Only consider figure 3 for part (a) & part (b).

CO2	(a)	**Calculate** the maximum fanout when the DTL circuit is connected to identical loads, the same as the driver circuit in figure 3 and the inputs are set to $A = B = 5V$.	[8]
CO2	(b)	**Calculate** the total power dissipation of the DTL circuit in figure 3 when $A = 5V$ & $B = 0V$.	[5]

The designer now wants to make an AND gate by using the DTL-NAND gate (figure: 3) as the driver and the RTL-NOT gate (figure 4) as the load.

CO2	(c)	**Calculate** the maximum number of RTL NOT gates a single DTL NAND gate can drive when the input of the DTL is both high ($A = 5V$, $B = 5V$).	[7]

Question 4 **[20]**

For the RTL circuit in Figure 5, assume $V_{BE}(saturation) = 0.8V$, $V_{CE}(saturation) = 0.2V$, $V_\gamma(transistor) = 0.5\ V$, $\beta_F = 30$, $V_{OH} = 10\ V$, $V_{OL} = 0.2\ V$

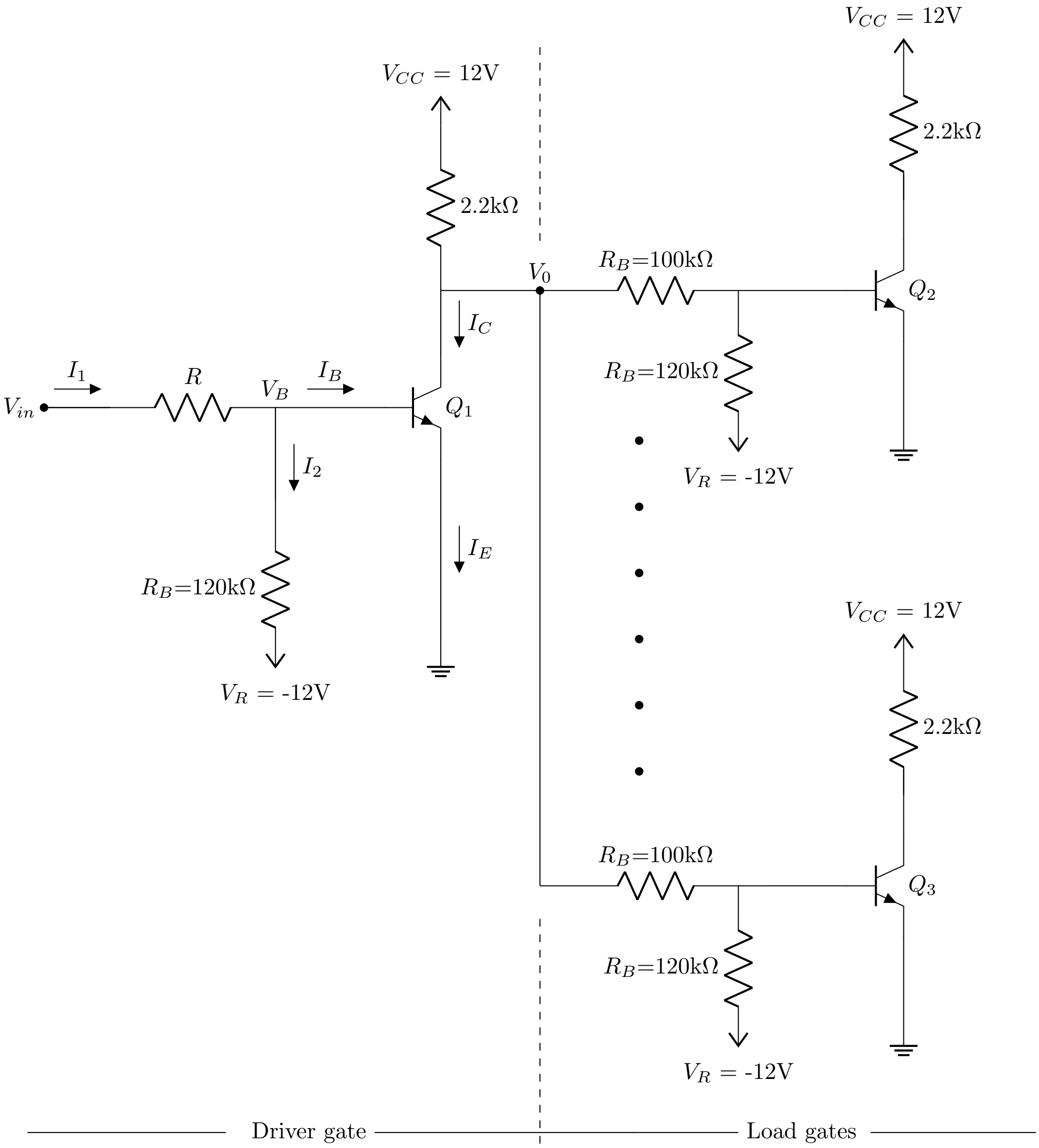

Figure 5

CO2	(a)	Consider no loads are attached to the driver circuit, **determine** the total power dissipated in the driver when input, $V_{in} = 0.2$V [*Assume* R = 10kΩ].	[4]
CO2	(b)	For high noise margin (V_{NH}) of 6V, **estimate** the value of R in kΩ when no loads are connected to the driver.	[8]
CO2	(c)	**Determine** the maximum possible fanout [*Assume* R = 10kΩ].	[8]